

Is the 360° street view redundant?

Phase-2 attention pruning — Qwen3.6-27B, $k=1, 2, 3$, $n=1663$ solved questions

The question

Each urban VQA question shows **4 street-view cameras** (fwd, bwd, cross_left, cross_right).

Hypothesis

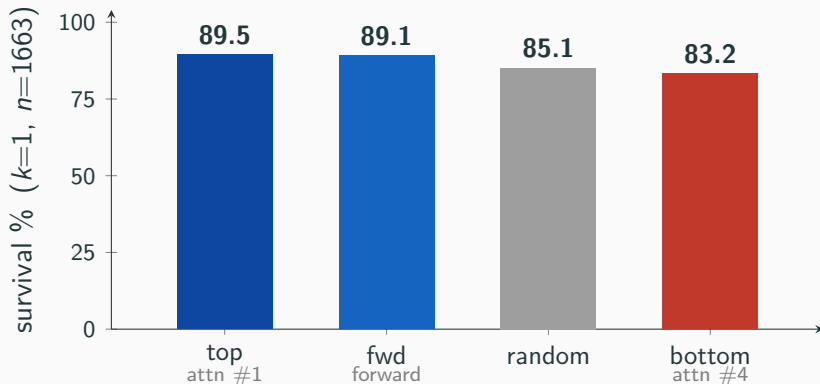
The full 360° view is **redundant**: the **one image the model attends to most** keeps the answer correct — and that image is the **forward camera**.

Method. Solve each question on all 4 images → keep only the k most-attended → re-ask greedily.

Survival % = fraction of solved questions still correct on the kept subset.

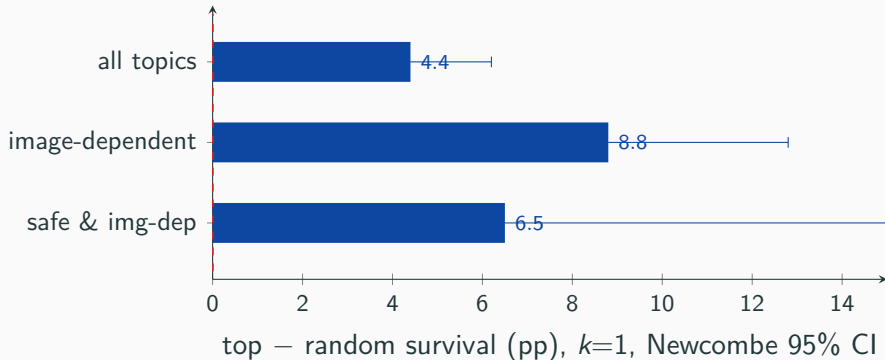
Arms: **top** (most-attended) | **fwd** (always forward) | random (pooled over 3 seeds) | **bottom** (least-attended).

One image is (almost) enough



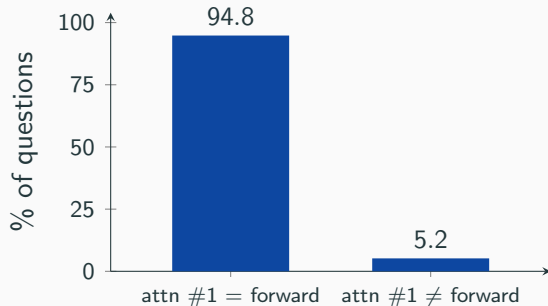
top $\approx$ **fwd** > random > **bottom**. The single most-attended image keeps the answer correct **9 times in 10** — the real evidence is the **gap over a random image**.

The gap over a random image is real — but small



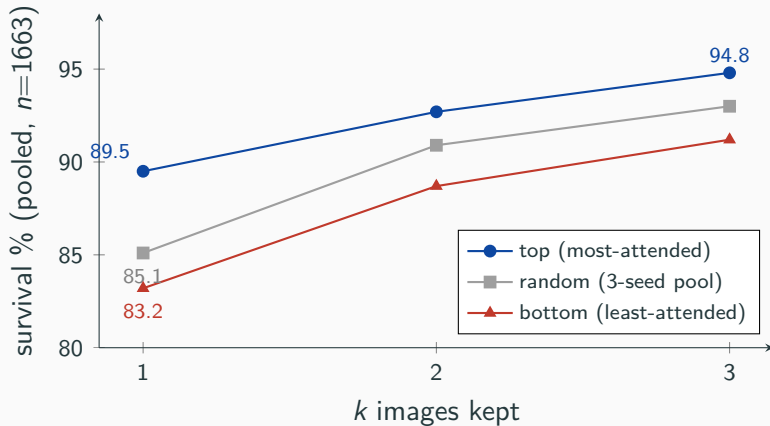
- Over **all topics**: **+4.4 pp** [+2.6, +6.1] — excludes 0.
- On the **image-dependent subset** (questions not answerable from priors): **+8.8 pp** [+4.8, +12.4] — the cleanest signal.
- On **leakage-safe topics only**, the all-pool gap CI *touches* 0 (+4.2 [-1.2, +8.9]): the headline rests on the image-dependent gap.

The attended image is the forward camera



- Attention's #1 image **is the forward camera in 94.8%** of questions.
- Pooled survival: **top** 89.5% $\approx$ **fwd** 89.1% — picking by attention and “always forward” land in the same place.
- On the 86 questions where attention *disagrees* with forward, it picks `cross_left/right` or `bwd` — and trends ahead of `fwd` (+9.3 pp, CI touches 0 at $n=86$).

More images help everyone — the curve



Top leads at every k , but the **gap narrows** as k grows (+4.4 $\rightarrow$ +1.8 $\rightarrow$ +1.8 pp): the redundancy is real — by $k=3$ which three hardly matters. The win is concentrated at $k=1$.

Per topic — forward beats random almost everywhere

topic	forward	random	gap	blind
road surface	97.5	92.5	+4.9	.96
junction type	91.2	91.1	+0.1	.61
transit density	90.8	86.2	+4.6	.30
road type	89.4	85.6	+3.8	.60
urban density	86.9	81.4	+5.5	.55
land use	87.1	82.3	+4.9	.52
building height	85.8	79.9	+6.0	.47
amenity richness	77.2	74.7	+2.5	.37
pooled	89.1	85.1	+3.9	

The **forward image** beats a **random** one in every topic. **Bold blind** ($< .45$) = leakage-safe; the rest survive partly from **class priors**, not vision — read those gaps with care.

- ✓ The 360° view is **redundant** — **one** street-view image gives 89.5% survival, and that image is the **forward camera** (attention's #1 in 94.8%).
- ✓ The right image **beats a random one** by **+4.4 pp** (CI [2.6, 6.1]) overall, **+8.8 pp** on image-dependent questions.
- ✗ Caveat: several topics carry **class-prior leakage**; on leakage-safe topics alone the raw gap touches 0. The **image-dependent** subset carries the claim (+8.8 pp).

3–4× fewer image tokens per question, same answer at $k=1$.

Repro: `python attention_prune/analyze_prune.py` | $k=1, 2, 3$, 3-seed random, $n=1663$.